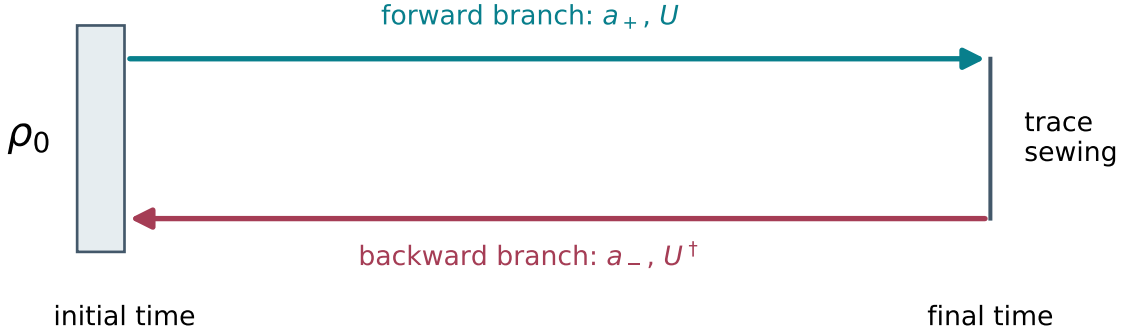
forward branch: a_+, U
ρ_0
trace
sewing
backward branch: $a_-, U^\dagger$
initial time
final time